

6.2 Configure EtherChannel

Configure EtherChannel

Configuration Guidelines

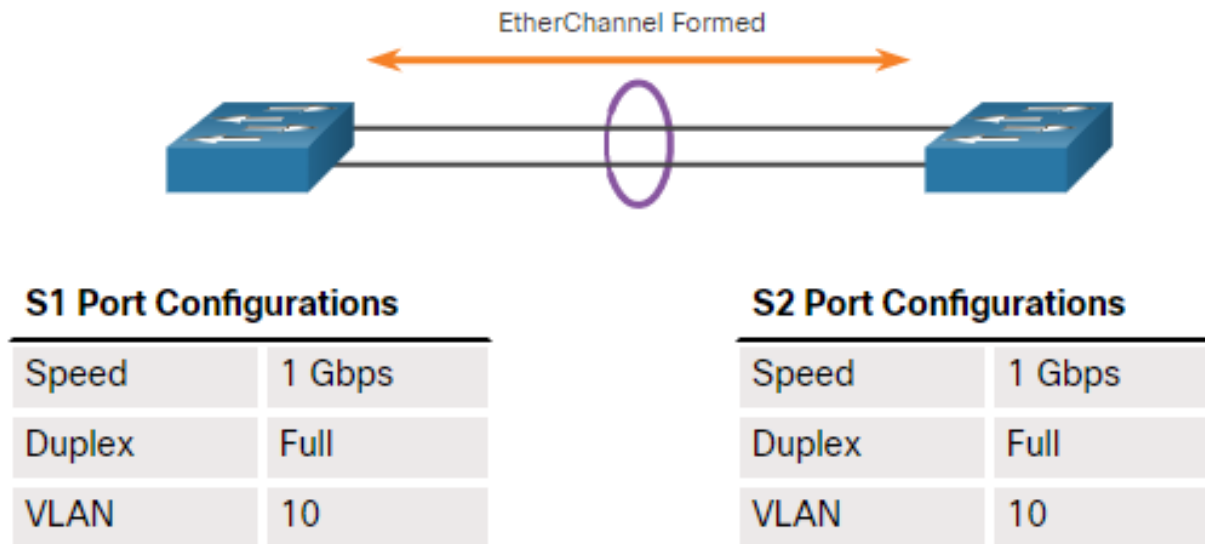
The following guidelines and restrictions are useful for configuring EtherChannel:

- **EtherChannel support** - All Ethernet interfaces must support EtherChannel with no requirement that interfaces be physically contiguous.
- **Speed and duplex** - Configure all interfaces in an EtherChannel to operate at the same speed and in the same duplex mode.
- **VLAN match** - All interfaces in the EtherChannel bundle must be assigned to the same VLAN or be configured as a trunk (shown in the figure).
- **Range of VLANs** - An EtherChannel supports the same allowed range of VLANs on all the interfaces in a trunking EtherChannel. If the allowed range of VLANs is not the same, the interfaces do not form an EtherChannel, even when they are set to **auto** or **desirable** mode.

Configure EtherChannel

Configuration Guidelines (Cont.)

- The figure shows a configuration that would allow an EtherChannel to form between S1 and S2.
- The port channel can be configured in access mode, trunk mode (most common), or on a routed port.
- If a Port channel is configured all interfaces are affected



Configure EtherChannel

LACP Configuration Example

Configuring EtherChannel with LACP requires the following three steps:

- **Step 1.** Specify the interfaces that compose the EtherChannel group using the **interface range** *interface* global configuration mode command. The **range** keyword allows you to select several interfaces and configure them all together.
- **Step 2.** Create the port channel interface with the **channel-group** *identifier* **mode** **active** command in interface range configuration mode. The identifier specifies a channel group number. The **mode** **active** keywords identify this as an LACP EtherChannel configuration.

```
S1(config)# interface range FastEthernet 0/1 - 2
S1(config-if-range)# channel-group 1 mode active
Creating a port-channel interface Port-channel 1
S1(config-if-range)# exit
S1(config-if)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 1,2,20
```

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LACP Configuration Example

Configuring EtherChannel with LACP requires the following three steps:

- **Step3.** To change Layer 2 settings on the port channel interface, enter port channel interface configuration mode using the **interface port-channel** command, followed by the interface identifier. In the example, S1 is configured with an LACP EtherChannel. The port channel is configured as a trunk interface with the allowed VLANs specified.

```
S1(config)# interface range FastEthernet 0/1 - 2
S1(config-if-range)# channel-group 1 mode active
Creating a port-channel interface Port-channel 1
S1(config-if-range)# exit
S1(config-if)# interface port-channel 1
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 1,2,20
```